

Effects of Tip Credit Elimination on Tipped Worker Earnings and
Employment in Washington D.C.

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Introduction

In recent years, the issue of tip credit elimination has continued to divide service workers and their would-be advocates. Currently, in the vast majority of US states, employers can take tip credit which allows them to pay their tipped employee a lower base hourly wage, provided that their total earnings per hour meet or exceed the standards set in relevant minimum wage laws. The federal minimum wage sits at \$7.25 per hour for non-tipped workers and \$2.13 per hour for tipped workers.

But while labor advocates petition governments to abolish tipped minimum wage in favor of a level playing field for all, in many instances, it is the tipped workers themselves arguing against the proposition. A fear of lower tips, layoffs, and longer working hours have placed many of these workers squarely on the side of their employers, who themselves argue that razor thin margins already only barely allow them to stay in business.

So why do policymakers continue to push for these changes if it seems that within the industry itself, many employers and employees actually agree? The 2024 Massachusetts Ballot Question 5; New Jersey Assembly Bill A5433; Flagstaff, Arizona's Proposition 414 - "The Minimum Wage Act"; and Washington D.C.'s Initiative 82 are all recent and current examples of the appetite of state and local governments (and voters) in abolishing tip credit. Are these jurisdictions simply operating on what sounds equitable or is there evidence to suggest that service workers are missing vital information?

While the contemporary literature would suggest tipped workers on the whole do not benefit from raising the tipped minimum wage, much of the research landscape utilizes similar state-level difference-in-differences comparisons to obtain similar results. This paper instead seeks to explore through county-level difference-in-differences comparisons using more recent

data via Washington D.C.'s Initiative 82 which was passed by Washington D.C. voters in 2022 and sought to phase out tip credit by 2027 (but was ultimately paused by the D.C. council in 2025). Using occupational, wage, and employment data from the areas of Washington D.C., Maryland, and Virginia, I determine the wage and employment effects of Initiative 82 in the D.C. area in a case study using data that includes the post-COVID-19 era. By using tipped workers in close-by counties of Maryland and Virginia as a control group and tipped workers in Washington D.C. as the treatment group, I find both positive wage effects and positive employment effects that challenge the prior literature.

Literature Review

David Neumark and Maysen Yen's paper on "The employment and redistributive effects of reducing or eliminating minimum wage tip credits" used state-level variation in minimum wages between 1990 and 2019. Through exploring quarterly administrative data from the Quarterly Census of Employment and Wages (QCEW) and the household survey data from the Current Population survey, they analyzed the employment effects, earning effects, and poverty/redistribution effects of raising the tipped minimum wage using difference-in-differences models with state and year fixed effects.

They found that each \$1 increase in the tipped minimum wage reduces employment among tipped restaurant workers by approximately 6.1%, corresponding to an employment elasticity of -0.08 (Neumark & Yen, 2023). While they do find small positive earnings effects among those still employed, ultimately these findings were statistically insignificant, contributing to their conclusion that any wage gains are largely offset by employment losses in

addition to potential tip erosion (Neumark & Yen, 2023). They were also unable to find any meaningful redistributive gains to low income families (Neumark & Yen, 2023).

Although Neumark and Yen assert that “[t]his evidence suggests that tipped restaurant workers do not benefit, on average, from increases in the tipped minimum wage” (Neumark & Yen, 2023), they state just before that they “find that the estimated effect on total earnings is negative (although indistinguishable from zero)” (Neumark & Yen, 2023). Neumark and Yen’s paper is seen as the seminal work in the study of tip credit, but with this slightly negative to ambiguous result, additional research using more recent data would help strengthen the fragile conclusions of the restaurant labor market’s relationship with increasing tipped minimum wage.

Much of Neumark and Yen’s work builds upon William Even and David Macpherson’s foundational study, “The Effect of the Tipped Minimum Wage on Employees in the U.S. Restaurant Industry”. Utilizing administrative data from QCEW (which they subsequently provided to Neumark and Yen for their analysis), they created a difference-in-differences model to estimate how tipped minimum wage increases affect employment in the restaurant industry. By comparing full-service restaurants to limited service ones, Even and Macpherson attempted to link tipped minimum wage changes to employment outcomes via a causal effect.

They found that a 10% increase in tipped minimum wage led to a 0.3-1.4% increase in average weekly earnings and 0.7-0.8% employment decrease (Even & Macpherson, 2014). Similar to Neumark and Yen, their wage effect findings were not statistically significant while their employment effect findings were. They acknowledge “the small increase in the average wages may mask larger increases for the subset of workers that receive tips” (Even & Macpherson, 2014), but emphasize that even for specifically tipped workers, the effects are modest. While they “believe the empirical analysis provides convincing evidence that higher

tipped minimum wages increase earnings but reduce employment” (Even & Macpherson, 2014), the statistical insignificance of the earnings effect combined with the size of each of the effects once again do not allow for a clear conclusion on net welfare impact.

Even and Macpherson’s assertion that tipped minimum wages act similarly to regular minimum wages through increasing earnings and reducing employment opens the door to further explore recent developments in minimum wage studies to compare in parallel. Arindrajit Dube’s “Impacts of minimum wages: review of the international evidence” lends evidence that further research in this field, especially using local comparisons, is necessary to understand the current relationship. Dube argues that country-level and even state-level comparisons are unable to properly control for spatial heterogeneity. He finds that across states, “employment effects are small up to around 59% of the median wage” (Dube, 2019) and that across sub-state county-level variation, “employment effects are small up to around 81% of the median wage” (Dube, 2019). Dube (2019) asserts that many of the studies that compare minimum wage between states using difference-in-differences evaluations assume the “parallel trends” assumption: “that absent the intervention, the outcomes in the treatment and control groups would have evolved in a parallel fashion” (Dube, 2019). But he finds support in favor of stronger parallel trends in contiguous areas rather than across whole states due to the likelihood of adjacent areas to experience similar economic shocks (Dube, 2019). While comparing states to one another could be a spatial heterogeneity problem, using Washington D.C. and its surrounding areas of Virginia and Maryland can better control for this issue and provide more meaningful results.

Additionally, Dube points to many examples where the gradual phase-in of minimum wage allows for employment to slowly adjust rather than experiencing a single large shock. Through a gradual approach, more localized comparisons, and more recent data, this paper

explores Washington D.C. Initiative 82 phase-in to see if Dube's minimum wage results extrapolate to tipped minimum wage.

Data

To test this, the following section describes the data used to construct the empirical analysis. The data employed in this study comes from the Bureau of Labor Statistics (BLS) Quarterly Census of Employment and Wages (QCEW). The QCEW is derived from unemployment insurance tax records that all employers are legally required to file and it covers approximately 95% of all U.S. employment. This means that the QCEW is not a sample-based survey but rather a near complete census of establishments that are covered by unemployment insurance. The QCEW is kept up to date via quarterly reports regularly submitted by employers on employment levels and wages paid which are then aggregated by the BLS on a county and industry level. By using the whole population of data instead of a sample, I am able to eliminate sampling error and non-response bias.

From the QCEW, this paper utilizes quarterly employment and wage statistics for the full-service restaurant industry in Washington D.C. as well as all five neighboring counties (Montgomery County, MD; Prince George's County, MD; Arlington County, VA; Fairfax County, VA; Alexandria City, VA) and four additional counties (Charles County, MD; Loudoun County, VA; Howard County, MD; Anne Arundel County, MD) in close proximity (two small commuter jurisdictions were excluded due to data irregularities). The additional counties were selected by proximity to Washington D.C., in line with Dube.

The unit of observation in this dataset is county-quarter. Each observation represents the full-service industry in a specific county during a specific quarter. They include only the

employment of non-federal government employed workers. Observations were collected within the timeframe of quarter 1 in 2017 to quarter 2 in 2025 and include average weekly wages, quarterly employment levels, and total wage bills for each county-quarter. The pre-treatment period is quarter 1 in 2017 to quarter 4 in 2022, before the enactment of Washington D.C.'s Initiative 82 whereas the post-treatment period is quarter 1 in 2023 to quarter 2 in 2025. This provides a solid foundation for testing parallel trends to strengthen the causality of the treatment.

There are weaknesses to the data too, however. Tip reporting is inconsistent and is often lower than real tips which can warp wages in unknown ways. Since this data is de-identified to maintain confidentiality, there is no ability to examine individual worker data, distributional effects, or worker population composition. And, while county-level data is quite granular, it still masks within-county variation.

The QCEW, while the same source that Neumark & Yen and Even & Macpherson used, is being used in this paper to specifically examine D.C.'s Initiative 82 rather than broader state-level policies. With the use of D.C. in particular, I am able to compare its policy to regional neighbors, potentially strengthening the parallel trends assumption relative to studies comparing non-contiguous regions. This data is also more recent than the data that is used in other research.

Table 1: Descriptive Statistics

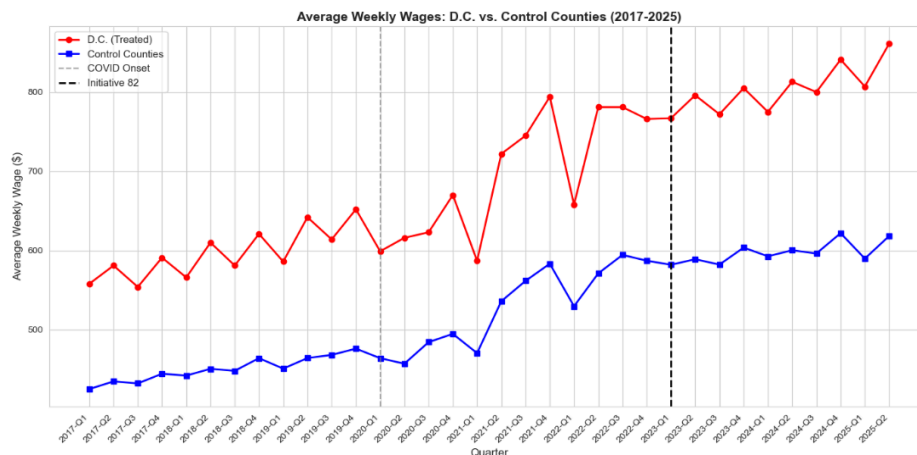
	avg_wkly_wage	avg_employment	qtrly_estabs_count	total_qtrly_wages	oty_avg_wkly_wage_pct_chg	oty_month1_emplvl_pct_chg
Count	340	340	340	340	340	340
Mean	\$538.04	10,232.829	424.576	\$76,920,840.33	4.642%	3.948%
Std	\$100.40	7407.224	302.484	\$68,753,137.05	5.337%	28.177%
Min	\$341.00	1158.333	72.000	\$5,316,688.00	-5.600%	-77.100%
Max	\$861.00	30749.667	1149.000	\$339,197,533.00	24.900%	143.600%

Table 2: Variable Definitions

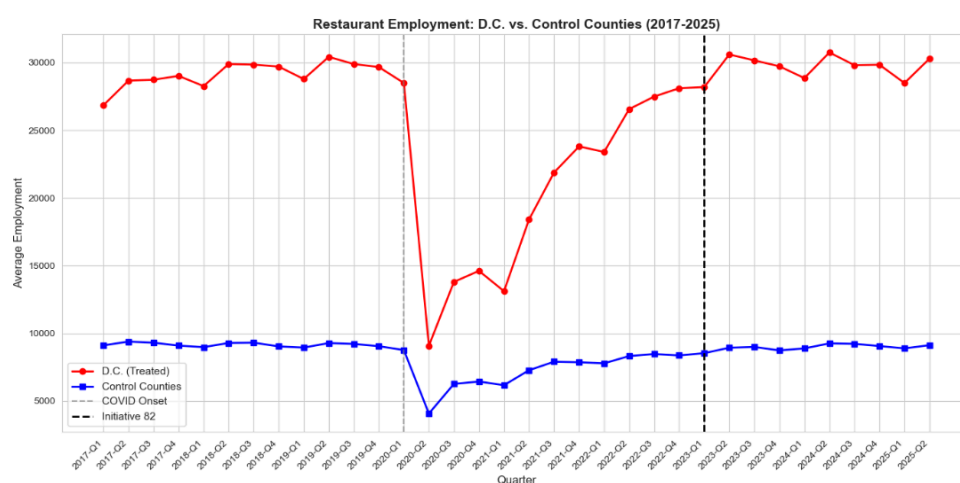
Column	Description	Unit	Role
avg_wkly_wage	Average Weekly Wage (\$)	Dollars/Week	Outcome #1
avg_employment	Average quarterly employment (jobs)	Count	Outcome #2
qtrly_estabs_count	Number of restaurant establishments	Count	Descriptive
total_qtrly_wages	Total quarterly wage bill (\$)	Dollars	Descriptive
oty_avg_wkly_wage_pct_chg	Year-over-year wage growth (%)	Percent	Trend analysis
oty_month1_emplvl_pct_chg	Year-over-year employment growth (%)	Percent	Trend Analysis

The dataset contains 340 observations and captures the quarterly labor market conditions in full-service restaurants. Average weekly wages range from \$341 to \$861 with a mean of \$538.04, while average quarterly employment spans from 1158 to roughly 30,750 workers with a mean of 10,233 workers, which reflects substantial variation in market size. Year-over-year wage growth averages 4.642% with moderate dispersion, whereas the year-over-year employment level change is more volatile with a standard deviation of 28.177% and a range from -77.1% to 143.6%, capturing pandemic-era disruptions that warrant consideration in the empirical analysis.

Additionally, a preliminary exploration of the data affords several broad tendencies in the data to discuss:



Through analyzing average weekly wages, we can see that border counties do indeed exhibit co-movement pre-treatment. While slightly less clear pre-COVID-19, during COVID-19 makes it clear that average wages do move together in these counties under common shocks. However, post-2023, Washington D.C.'s wage premium over control counties appears to widen slightly, suggestive of a treatment effect. Also shown is the strong cyclicalty of wages with D.C. and the weaker but noticeable cyclicalty in control counties as well as the fact that D.C. generally has higher wages than its surrounding counterparts.



Similarly, a close examination of employment trends yields comparable parallel trends, particularly pre-COVID-19 and then also during COVID-19. Even slight shocks to employment reflect in both Washington D.C. and the control counties. Employment levels, however, are less cyclical than wages which could be attributed to the fact that hiring and firing employees is expensive and tips are higher during summer and holidays. Additionally, it is clear that wages were able to withstand COVID-19 better and bounce back quicker than employment levels. These preliminary trends motivate the two-way fixed effects framework described in the following section.

Econometric Model

The hypothesis of the author is that the gradual increase in tipped minimum wage in Washington D.C. through Initiative 82 resulted in increased weekly wages and had no negative effect on employment. This is because, while traditionally it is held that increasing minimum wage results in negative employment effects, for labor markets, employers are actually observed to have monopsonistic wage-setting power. Under monopsony, employment is often constrained not by a firm's demand for labor but rather the supply of labor to the firm. Minimum wage increases could therefore encourage low-wage workers to stay with firms which can reduce vacancies, allow employers to fill open positions with the unemployed instead of job hoppers, and increase employment (Dube, 2019).

My research topic exists within the intersection of Dube (2019)'s work on localized monopsonistic minimum wage and Neumark & Yen (2023) and Even & Macpherson (2014)'s work on state-level tipped minimum wage analysis. While Neumark & Yen (2023) and Even & Macpherson (2014) found insignificant positive earning effects and negative employment effects as a result of increasing tipped minimum wage, Dube (2019) posits that state-level analyses are unable to properly control for spatial heterogeneity. While his work specifically applies to minimum wage, not tipped minimum wage, current literature does not address if, given localized controls, Dube (2019)'s claims that increasing minimum wage actually has little to no effect on employment could be extrapolated to tipped minimum wage. In addition to this novel intersection and being more granular than current literature, this study also incorporates labor market data during and after COVID-19.

Dube argues that county-level comparisons like the one in this study better represent the real cost of increasing minimum wages on unemployment, unlike the state-level comparisons of

Neumark & Yen and Even & Macpherson. Actual tipped minimum wage levels are used rather than a binary post variable to allow the model to more accurately represent the functionality of the gradual Initiative 82. Additionally, using data from Q1 2017 to Q2 2025 gives a substantial pre-treatment period for parallel trends testing and enough post-treatment quarters to observe phase-in effects meaningfully.

To test my hypothesis, I employ the following econometric equations:

Model 1: Average Weekly Wages

$$\ln(\text{avg_wkly_wage_it}) = \alpha + \beta(\text{tip_min_it} \times \text{dc_i}) + \mu_i + \delta_t + \varepsilon_it$$

Model 2: Employment Levels

$$\ln(\text{avg_employment_it}) = \alpha + \beta(\text{tip_min_it} \times \text{dc_i}) + \mu_i + \delta_t + \varepsilon_it$$

avg_wkly_wage_it is the average weekly wage for a county *i* in period *t*.

avg_employment_it is average quarterly employment for county *i* in period *t*. *tip_min_it* is the tipped minimum wage for county *i* in period *t*, constructed by subtracting county *i*'s pre-2023 baseline tipped minimum wage and *dc_i* is a D.C. dummy and time-invariant. μ_i represents county fixed effects while δ_t represents time fixed effects.

Instead of OLS, I've decided to use a two way fixed effects model to represent the changes of the treatment in both average weekly wages and employment levels. This allows the model to absorb all time-invariant differences between D.C. and the control counties through county fixed effects as well as the common shocks that affect all counties simultaneously through time fixed effects, addressing omitted variable bias. It also provides the model with a difference-in-differences estimator that is able to isolate only the variation that is specific to D.C. and the post-treatment period.

The dependent variables are both transformed using natural log to increase interpretability, comparability, and help normalize the distributions which are right skewed. While some consideration was made to use population-weighted OLS where each observation is weighted by the county's annual Census population estimate, ultimately this was dropped as it produced unrealistically high employment-to-resident-population ratios in smaller counties with workforces largely composed of commuters who live elsewhere. Given the substantial variation in market size across counties, homoscedasticity is unlikely, and county-clustered standard errors provide heteroscedasticity-consistent inference in addition to accounting for serial correlation, although the small cluster count ($n=10$) is a limitation.

While variance inflation factors confirm that multicollinearity is not a primary concern in this specification, as key variables are constructed to capture distinct sources of variation, there is still potential concern for simultaneity bias. Initiative 82 was passed by voter referendum, but the pace of implementation was subsequently modified by the D.C. Council, suggesting the phase-in schedule was not fully exogenous to labor market conditions.

Results

Table 3: Log Weekly Wage Estimates Across Specifications

	Main Model	RC1: Binary DiD	RC2: No COVID	RC3: Contiguous Only
Log Weekly Wage	0.0073*** (0.0019)	0.0202*** (0.0050)	0.0065*** (0.0021)	0.0082*** (0.0017)
Observations	340	340	220	204
Within R ²	0.0101	0.0096	0.0118	0.0192
Entity FE	Yes	Yes	Yes	Yes
Time FE	Yes	Yes	Yes	Yes

*Standard errors clustered at the county level in parenthesis. Significance is reported for the following levels: * $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$*

Table 4: Log Quarterly Employment Estimates Across Specifications

	Main Model	RC1: Binary DiD	RC2: No COVID	RC3: Contiguous Only
Log Quarterly Employment	0.0251*** (0.0033)	0.1024*** (0.0117)	0.0014 (0.0052)	0.0306*** (0.0032)
Observations	340	340	220	204
Within R ²	0.0129	0.0191	0.0012	0.0211
Entity FE	Yes	Yes	Yes	Yes
Time FE	Yes	Yes	Yes	Yes

Standard errors clustered at the county level in parenthesis. Significance is reported for the following levels: * $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$

Running the difference-in-differences regressions on the two models, I found that a \$1 increase in the tipped minimum wage is associated with a +0.73% increase in average weekly wages for the full-service industry, statistically significant at the 1% level ($p < 0.01$). I also found that a \$1 increase in the tipped minimum wage is associated with a +2.51% increase in average quarterly employment for the full-service industry, statistically significant at the 1% level ($p < 0.01$). Three robustness checks largely validate these results. RC1 treats the treatment as a single shock rather than a gradual phase in, RC2 removes the time period in which COVID-19 was considered a public health emergency, and RC3 uses only regions that are contiguous with Washington D.C. in the regression. While all of the models show significant and positive changes in log wage in table 3, the changes in log employment lose significance when excluding the COVID-19 time period in table 4. This suggests that the positive employment effect may be partially driven by post-pandemic labor market recovery coinciding with the Initiative 82 phase-in, rather than the policy alone.

While the within R-squared for both main models are modest (0.0101 for wages, 0.0129 for employment), this makes sense as two way fixed effects models normally see most of the variation absorbed by the fixed effects. Additionally, the pooled F-test of 547.81 ($p = 0.0000$) for

log wage and 1061.6 ($p=0.0000$) for log employment justifies the panel structure of this study versus pooled ordinary least squares.

These results generally support my hypothesis: due to Initiative 82 in Washington D.C., wages increased and employment showed no negative effect. As shown before, there is some concern when identifying a causal policy effect without the COVID-19 period since the resulting coefficient on log employment is not significant, but the evidence across specifications of this study suggests no large negative effect on employment.

This result was consistent with a gradual phase-in approach together with localized controls for spatial heterogeneity per Dube (2019), showing that his arguments can apply to tipped minimum wage in addition to regular minimum wage. These findings agree with current literature in finding positive wage effects (although this study finds that they are significant), but they diverge from Neumark & Yen (2023) and Even & Macpherson (2014) as, while they found significant negative effects on employment (-6.1% and roughly 0.8% per \$1 increase in tipped minimum wage respectively), this study found significant positive effects on employment (+2.51% per \$1 increase in tipped minimum wage). This suggests that more research must be done to explore the monopsonistic effects of the restaurant industry on particularly tipped minimum wage as this case study refutes present understanding of the consequences of tip credit elimination. It also suggests that research on this topic has been subject to methodological limitations and must generally be done on a more minute level to better control for regional differences.

Based on these findings, I would recommend policymakers to consider gradual tip credit elimination phase-ins due to its positive effects on wages and its lack of a negative effect on employment. Gradual phase-in allows firms to adjust pricing, staffing, and other operational

costs incrementally without having to resort to large-scale layoffs or other negative consequences, consistent with Dube's (2019) monopsonistic mechanism in which minimum wage increases correct employer wage-setting power rather than distort competitive equilibrium. Neumark & Yen (2023) and Even & Macpherson's (2014) work appears to overstate the traditional disemployment effects of raising tipped minimum wage in more localized contexts with gradual implementation, particularly in dense urban areas where monopsonistic labor market effects are strongest. Policymakers in these areas should view the results of Washington D.C.'s Initiative 82 as a positive example of the benefits of tip credit elimination and D.C. Council itself should consider unpausing the wage increase schedule. However, further research into this topic is still encouraged due to the aforementioned loss of significance of employment effects when excluding the COVID-19 period. This study cannot conclusively isolate the policy effect from the post-pandemic labor market recovery.

Conclusion

Despite these limitations, the weight of evidence across specifications points toward a meaningful policy effect. This paper sought to understand the wage and employment effects that occurred in Washington D.C. for full-service industry workers after the passage of Initiative 82 which outlined a plan to eliminate tip credit gradually until 2027. Through the implementation of two TWFE models, I was able to estimate these effects holding time-invariant and county-level shocks constant. This paper found that a \$1 increase in the tipped minimum wage resulted in a 0.73% increase in the average weekly wage and a 2.51% increase in the average quarterly employment in Washington D.C., with both results being highly significant.

These findings challenge the contemporary tipped minimum wage standard of Neumark & Yen and Even & Macpherson using a more localized approach that was consistent with Dube's inquiry into employment effects due to minimum wage increases. While the results were highly significant, it should be noted that the model lost significance when excluding the COVID-19 period. This study also potentially suffers from simultaneity concerns since the D.C. Council's modification of the phase-in schedule suggests that the implementation of this policy is not fully exogenous to labor market conditions.

Because of how this paper challenges the current status quo, future research using similar methodology in different areas and different time periods is strongly encouraged. Replicating this analysis in other cities with gradual phase-ins of tip credit elimination would further strengthen this study and provide additional insight on how this policy truly affects wages and employment. This work could also be extended if individual worker-level data becomes accessible to examine distributional effects. But, for now, this paper recommends that policymakers (like those currently debating in Massachusetts, New Jersey, or Flagstaff) pursue tip credit elimination as it increases wages and has no negative effect on employment.

References

- Even, William E., and David A. Macpherson. "The effect of the tipped minimum wage on employees in the U.S. restaurant industry." *Southern Economic Journal*, vol. 80, no. 3, Jan. 2014, pp. 633–655, <https://doi.org/10.4284/0038-4038-2012.283>.
- Dube, Arindrajit. "Impacts of Minimum Wages: Review of the International Evidence." *Independent Report*, UK Government, 2019.
- Neumark, David, and Maysen Yen. "The employment and redistributive effects of reducing or eliminating minimum wage tip credits." *Journal of Policy Analysis and Management* 42.4 (2023): 1092-1116.
- United States, Department of Labor, Bureau of Labor Statistics. "Quarterly Census of Employment and Wages: Full-Service Restaurants (NAICS 722511), 2017–2025." QCEW Data Files, 19 Feb. 2026, www.bls.gov/cew/downloadable-data-files.htm.